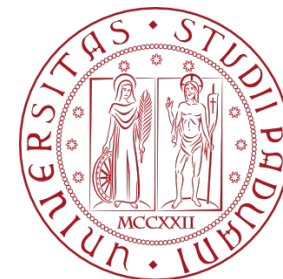


Sentiment Analysis



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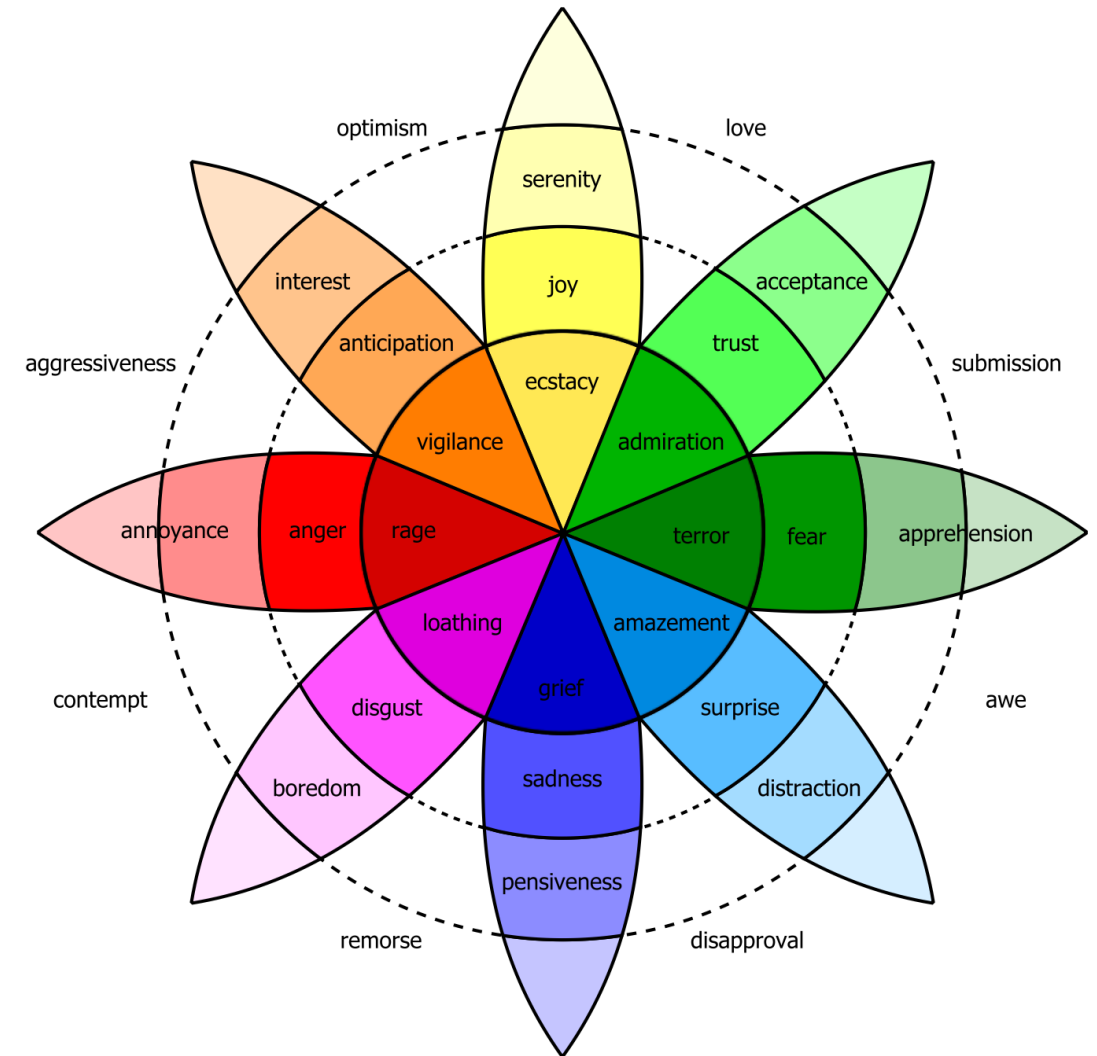
- Identify the sentiment of a text can be useful for
 - Determining the attitude of people towards a brand, a law, a product, a person..
 - Much like conducting “polls” – this is very popular in social science
 - It can be useful to analyse a person’s mood: looking for early depression signs from social media etc...
- Sentiment can be a component for downstream applications
 - Ex. summarizing meetings or finding parts of a conversation where people are especially excited or engaged
 - Detect frustration in an interaction between a human and a conversational agent

- Detect the personality of a user (ex. extrovert) to adapt a conversational agent
- Here we focus on certain words, ones that carry particularly strong cues to sentiment
 - Instead of letting the learning algorithm devise the important words/features, we inject our knowledge about them into the system

Theory of Emotions



- Theories are the 6 emotions (Elkman), supposedly present in all cultures:
 - surprise, happiness, anger, fear, disgust, sadness
- Plutchik theory has 8 emotions in 4 opposing pairs
- Other works identify 3 axis:
 - valence: the pleasantness of the stimulus
 - arousal: the intensity of emotion provoked by the stimulus
 - dominance: the degree of control exerted by the stimulus



- Binary lexicons representing the sentiment (valence) of words
 - General Inquirer (1967)
 - 1915 positive words, 2291 negative words
 - MPQA Subjectivity lexicon (2005)
 - 2718 positive words, 4912 negative words, and a list of subjective words and phrases
- Lexicons considering further aspects
 - The NRC Word-Emotion Association Lexicon, EmoLex (2013), gives binary labels to 14k words according the Plutchik schema

- LIWC, Linguistic Inquiry and Word Count, is a widely used set of 73 lexicons containing over 2300 words (2007)..
 - In addition to sentiment-related lexicons it includes lexicons for categories like anger, sadness, cognitive mechanisms, inhibition

Positive Emotion	Negative Emotion	Insight
appreciat*	anger*	aware*
comfort*	bore*	believe
great	cry	decid*
happy	despair*	feel
interest	fail*	figur*
joy*	fear	know
perfect*	griev*	knew
please*	hate*	means
safe*	panic*	notice*
terrific	suffers	recogni*

* Means all suffixes are included in the lexicon

- Non Binary lexicon
 - The NRC Valence, Arousal, and Dominance - VAD - (2018) assigns valence, arousal, and dominance scores to 20,000 words.

Valence		Arousal		Dominance	
vacation	.840	enraged	.962	powerful	.991
delightful	.918	party	.840	authority	.935
whistle	.653	organized	.337	saxophone	.482
consolation	.408	effortless	.120	discouraged	.0090
torture	.115	napping	.046	weak	.045

- valence is the pleasantness of a stimulus, arousal its intensity, dominance is the degree of control exerted by a stimulus
- How to annotate it?

- Non Binary lexicon
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- Annotation: ask annotator to select best/worst (ex. according to sentiment) among four terms. The score of a term is the percentage of times it has been best – percentage of times it has been worst

- Lexicons might need to be adapted to the current domain, culture or context
 - Small is positive for smartphones (is it still?), negative for apartments
- Hence the need to learn / fine tune them from data
- Semantic Axis:
 - define positive and negative vectors in a word embedding space
 - Define the axis between such vectors (the vector connecting them)
 - project words into the axis and determine whether the positive or the negative is closer

- The positive and negative vectors are determined on the basis of a set of initial seed words
- Words having different sentiment in different context can be left to the learning algorithm to deal with
- Different seed words can be provided for each context

Domain	Positive seeds	Negative seeds
General	good, lovely, excellent, fortunate, pleasant, delightful, perfect, loved, love, happy	bad, horrible, poor, unfortunate, unpleasant, disgusting, evil, hated, hate, unhappy
Twitter	love, loved, loves, awesome, nice, amazing, best, fantastic, correct, happy	hate, hated, hates, terrible, nasty, awful, worst, horrible, wrong, sad
Finance	successful, excellent, profit, beneficial, improving, improved, success, gains, positive	negligent, loss, volatile, wrong, losses, damages, bad, litigation, failure, down, negative

- Given a set of embeddings for positive (negative) seed words, we compute $\mathbf{V}^+$ ($\mathbf{V}^-$) the vector representing positive (negative) sentiment

$$\mathbf{V}^+ = \frac{1}{n} \sum_1^n E(w_i^+)$$

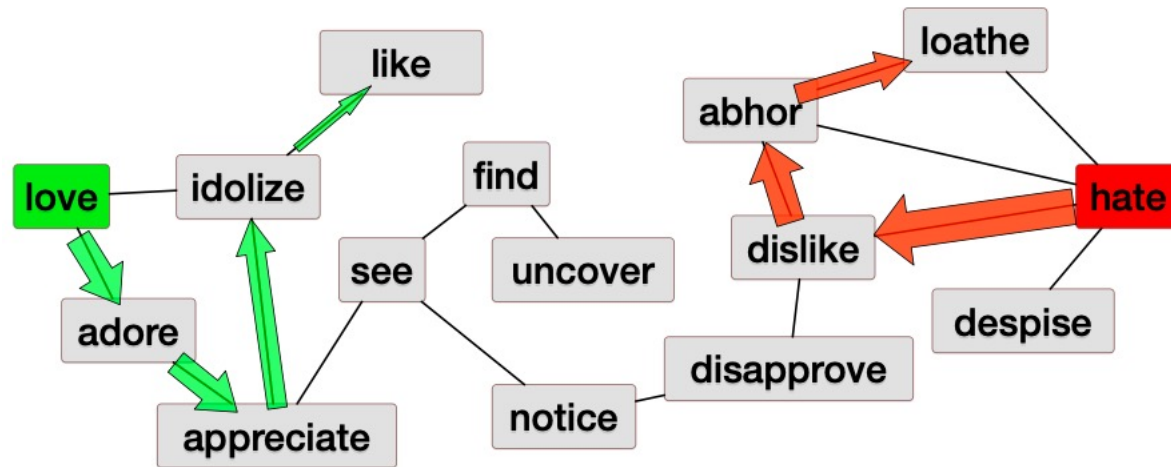
- The Axis is defined as

$$\mathbf{V}_{axis} = \mathbf{V}^+ - \mathbf{V}^-$$

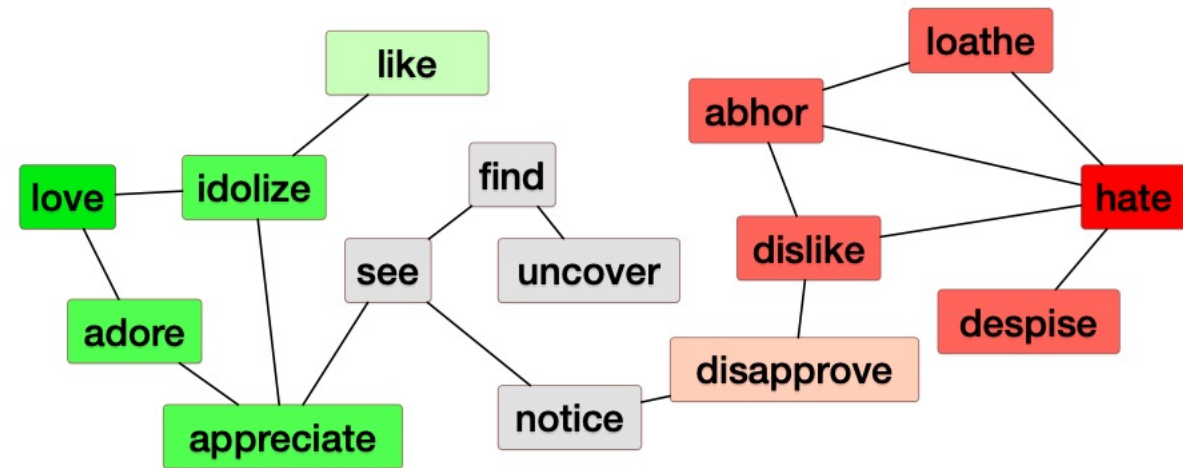
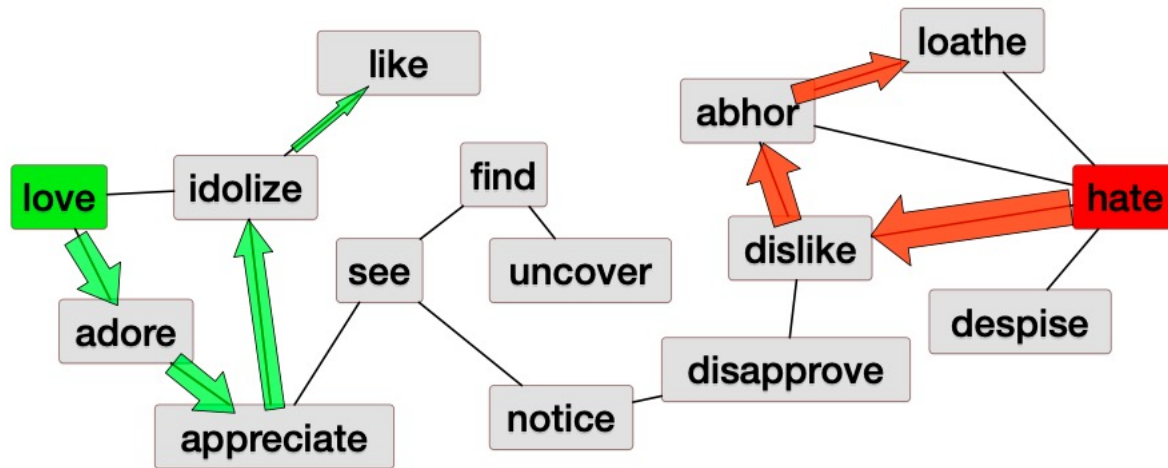
- And a new word is scored as

$$\begin{aligned} \text{score}(w) &= (\cos(E(w), \mathbf{V}_{axis})) \\ &= \frac{E(w) \cdot \mathbf{V}_{axis}}{\|E(w)\| \|\mathbf{V}_{axis}\|} \end{aligned}$$

- Define a graph: one node per word, an edge for each of the k most similar words (weight the edges according to the same similarity function)
- Define seed words
- Propagate polarities on the graph from seed words
 - Perform random walks starting from each node and moving to a neighbouring node with probability proportional to the edge weight



- Score the nodes in the graph: count the percentage of times a node has been reached from any walk starting from a positive labelled node w.r.t. the total number it has been visited



- Confidence values for the scores can be obtained by repeating the walks from different seed sets

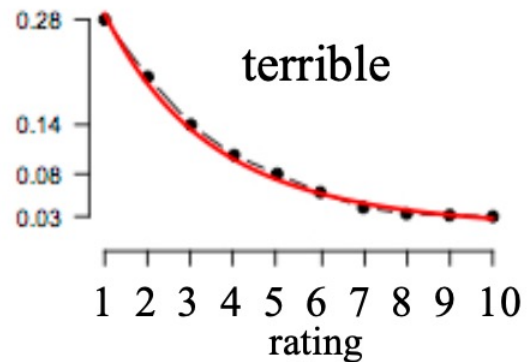
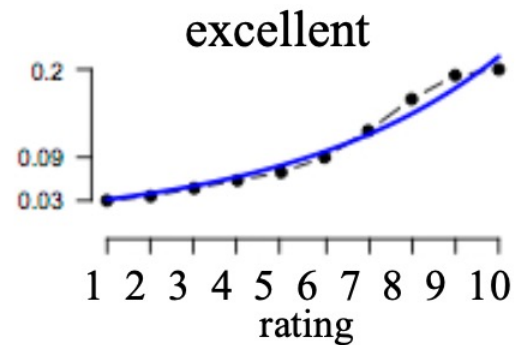
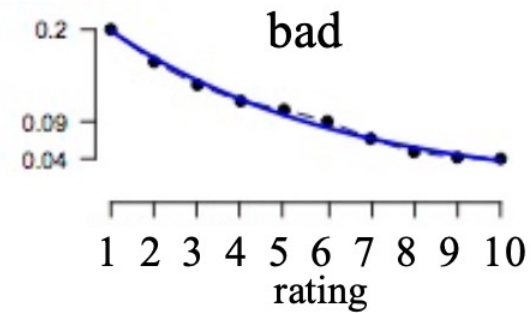
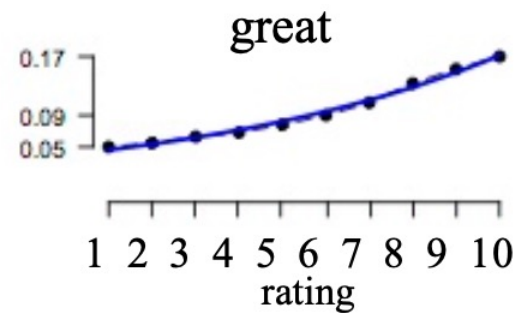
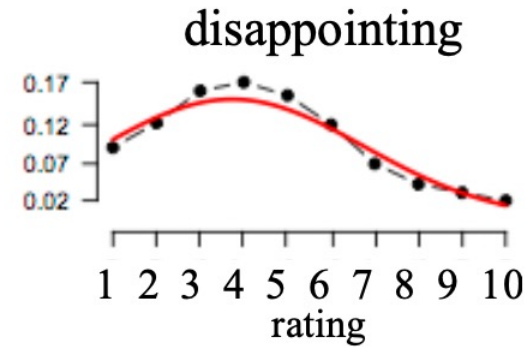
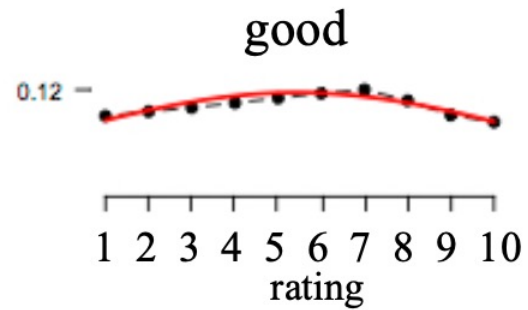
- Label Propagation has applications in various tasks, for example identifying online communities with different opinions on a topic
- Similarity measures other than cosine have been used:
 - Syntactic cues: two adjectives are considered similar if they were frequently conjoined by *and* and rarely conjoined by *but*
 - Adjectives with similar roots with one using morphological negation
 - Use a thesaurus to compute words that are similar to a positive one and that are similar to a negative one

- If we use reviews (for example with a 10 star rating), we can
 - Check the statistics of words appearing in 10 star reviews only (positive)
 - Check the statistics of words appearing in 1 star reviews only (negative)
- We can also observe the distribution over all the ratings

$$P(w|c) = \frac{\text{count}(w, c)}{\sum_{w \in C} \text{count}(w, c)}$$

$$\text{PottsScore}(w) = \frac{P(w|c)}{\sum_c P(w|c)}$$

Supervised Learning of Lexicons



- Lexicons can be used to extract features, for instance for a document:
- Binary features, i.e. 1 if there is a word in the document which is also in the positive (negative) lexicon
- Number of words of the lexicon appearing in the document

$$f_L = \sum_{w \in L} count(w)$$

- Weighted count of the words appearing in the lexicon (if scores available)

$$f_L = \sum_{w \in L} \theta_w^L count(w)$$

- What about assessing the sentiment towards an entity in a document (instead of the document itself)?
- We need to extract the phrases / sentences in which an entity is mentioned
 - Semantic Role Labelling could be of help
- (Field and Tsvetkov (2019)) train a system for scoring contextualised word embeddings
 - Get all contextualised word embeddings for a word, compute the average E
 - Get the VAD scores for the word; train 3 systems to learn a mapping from E to these scores
 - At prediction time, use the trained systems to get the VAD scores

Entity-Based Sentiment Analysis



- Now we are able to get the VAD scores for each entity
- We can average the values for each mention of an entity

- An Example for entities in a movie

- Power corresponds to Dominance
- Sentiment corresponds to Valence
- Agency corresponds to Arousal

